

Name

- For the ODE $y'' - 6y' + 13y = e^{-2t}$

- Write down the complementary sol.

$y_c(t) =$

- Compute a particular sol.

$y_p(t) =$

- The Aug Mat for $y(0) = 2, y'(0) = 3$ is

1. For the ODE $y'' - 5y' - 14y = \sin(t)$

- 1.1. Write down the complementary sol.

$y_c(t) =$

- 1.2. Compute a particular sol.

$y_p(t) =$

- 1.3. The Aug Mat for $y(0) = 1, y'(0) = 4$ is

- For the ODE $y'' - 4y' + 9y = 3\sin(2t)$ with

In[1]:= **Expand** [$(x - 2)^2 + 9$]

Out[1]= $13 - 4x + x^2$

- Write down the complementary sol.

$$y_c(t) =$$

- Compute a particular sol.

$$y_p(t) =$$

- The Aug Mat for $y(0) = 2, y'(0) = -3$ is

$$=$$

2. For the ODE $y'' - 2y' + 5y = 2\cos(3t)$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Compute a particular sol.

$$y_p(t) =$$

- 2.3. The Aug Mat for $y(0) = -2, y'(0) = 7$ is

$$=$$